

Introduction to Statistical Methods - Mid-Semester 1 Exam Guide

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ISM Midsem 1 Preparation Hub

Course: AMLCS ZC418 Introduction to Statistical Methods. The repository calendar schedules the examination for 19 September 2026, 01:00 PM-03:00 PM. The practice mock in this package is deliberately 90 minutes and 30 marks. Scope: Contact Sessions 1-8.

Start with the [syllabus map and standalone mini guide](#). The [printable exam guide](#) combines that guide with all three drill banks. Assume only high-school algebra and arithmetic.

The generated PDFs are not structurally tagged, so assistive-technology users should prefer these accessible Markdown sources. Direct administration of a PDF requires the externally supplied trusted seal; the repository intentionally does not invent or embed that seal.

Six-day plan

Day 1 - center, spread, and probability framing

Learn [Session 1](#), read the [Session 1 card](#), then attempt [C01-C03](#) and [N01-N03](#). Check every quartile convention and unit.

Day 2 - events, counting, and conditioning

Learn [Session 2](#) and [Session 3](#). Read the [Session 2 card](#) and [Session 3 card](#). Attempt conceptual [C04-C06](#) and [C07-C09](#), then numerical [N04-N06](#) and [N07-N09](#).

Day 3 - Bayes and naive Bayes

Learn [Session 4](#), read the [Session 4 card](#), then attempt the [Session 4 conceptual drills C10-C12](#) and [Session 4 numerical drills N10-N12](#). Label prior, likelihood, evidence, and posterior before calculating.

Day 4 - random variables and distributions

Learn [Session 5](#) and [Session 6](#). Read the [Session 5 card](#) and [Session 6 card](#). Attempt conceptual [C13-C15](#) and [C16-C18](#), then numerical [N13-N15](#) and [N16-N18](#).

Day 5 - sampling and integrated review

Learn [Session 7](#) and [Session 8](#). Read the [Session 7 card](#) and [Session 8 card](#). Attempt conceptual [C19-C21](#), [C25-C26](#), and [C22-C24](#). Attempt numerical [N19-N21](#), [N25-N27](#), and [N22-N24](#), then complete the [historical-source-aware practice](#). Items N25-N26 review Session 6 distributions; N27 and C25-C26 review Session 7 sampling and confidence intervals.

Day 6 - timed rehearsal and checking

Sit [Mock 01](#), or its [printable paper](#), for 90 minutes. After submission only: open the [answer key](#), [point rubric](#), or [solutions PDF](#). Rework lost marks, then finish with the [last-30-minutes route](#).

After attempting the banks only: use the gated [C01-C24 checking paths](#), [N01-N24 checking paths](#), [H01-H03 checking paths](#), and [additional C25-C26/N25-N27 explanations](#). These solution links are intentionally absent from Days 1-5.

Required answer form

For every calculation-bearing response, show the governing formula, supplied-value substitution, non-obvious intermediate steps, units or domain and validity conditions, and interpretation. For a qualitative applied response, explain any [not applicable](#) elements and still state conditions and interpretation.

ISM Midsem 1 Syllabus Map and Standalone Guide

The handout prescribes Sessions 1-8. The status columns keep prescribed scope separate from local teaching evidence.

Session	Handout-prescribed scope	Local teaching evidence
1	Center, variability, symmetry/skew, outliers, five-number summary, probability framing	Substantive slides
2	Probability axioms, exclusive and independent events, problem solving	Substantive slides
3	Conditional probability, independence, total probability	Substantive slides
4	Bayes proof and naive Bayes introduction	Substantive slides
5	Discrete/continuous random variables, moments, single and joint distributions	Substantive sampled 05 September recording
6	Bernoulli, binomial, Poisson, normal; introduction to t, F, chi-square	Prescribed; no substantive local teaching evidence used
7	Random and stratified sampling, sampling distributions, CLT, interval estimation	Prescribed; no substantive local teaching evidence used
8	Revision of covered topics	Prescribed review; no new theory

Prerequisite bridge

A set is a collection; $\cup$ means either event, $\cap$ means both, and $\emptyset$ means no outcomes. The complement of A means every outcome in the sample space that is not in A. $\sum$ means add across allowed values; an integral means continuous area across an interval. A factorial is $n! = n(n-1)\cdots 1$, with $0! = 1$. A square root reverses squaring. Probabilities are unitless and between 0 and 1.

Session 1: descriptive statistics and probability framing

Why it matters

A summary should reveal a typical value, spread, and shape without hiding unusual observations. Probability starts from a known model and reasons toward samples; statistical inference starts with sample information and reasons toward a population.

Core language and formulas

A population is the complete group; a sample is the observed part. A parameter such as population mean μ or variance σ^2 describes the population. A statistic such as sample mean $\bar{x}$ or variance s^2 is calculated from a sample.

Session 1 compact formulas

$$\bar{x} = \frac{\sum x_i}{n} \quad s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$\text{IQR} = Q_3 - Q_1 \text{ fences: } Q_1 - 1.5 \text{ IQR}, Q_3 + 1.5 \text{ IQR}$$

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2} \text{ for } k > 1 \text{ and } 0 < \sigma^2 < \infty$$

For values x_i with nonnegative weights w_i and positive total weight, the weighted mean is $\sum w_i x_i / \sum w_i$. The median is the middle ordered value, or the average of the two middle values. The mode is a most frequent value and need not be unique.

This guide uses the median-of-halves excluding the overall median quartile convention. Compute $\text{IQR} = Q_3 - Q_1$; the fences are $Q_1 - 1.5 \text{IQR}$ and $Q_3 + 1.5 \text{IQR}$. Fences are theoretical cutoffs. In a standard box plot, whiskers end at observed values within the fences, so they need not equal the fences.

The empirical rule is an approximation for an approximately normal distribution. Chebyshev says at least $1 - 1/k^2$ lies within k standard deviations for $k > 1$ when variance is finite and positive. It is a guaranteed lower bound. If variance is zero, every value equals the mean and the standardized formula is not used.

Pearson's second skew coefficient is $3(\bar{x} - \text{median})/s$. It is undefined when $s=0$; zero does not prove symmetry. Graphs such as histograms and box plots should be read alongside the number, because different shapes can share the same coefficient.

Checked illustration. Ordered times 2,4,5,7,8,14 have median $(5+7)/2=6$, $Q_1=4$, $Q_3=8$, $\text{IQR}=4$, and fences -2 and 14. The value 14 is on the fence and is not beyond it.

Session 2: probability axioms, events, and counting

An experiment has a sample space S of outcomes. An event is a subset. Relative to a die sample space, $\{4\}$ is elementary while $\{2,4,6\}$ is composite. The three axioms are nonnegativity, $P(S)=1$, and addition across countably many mutually disjoint events.

Session 2: probability and counting

$$P(A) \geq 0 \quad P(S) = 1 \quad \text{disjoint } A_i: P(\cup A_i) = \sum P(A_i)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(n,r) = \frac{n!}{(n-r)!} \quad C(n,r) = \frac{n!}{r!(n-r)!}$$

The complement rule is $P(\text{complement of } A) = 1 - P(A)$. Inclusion-exclusion subtracts the overlap once. Events are mutually exclusive when $A \cap B = \emptyset$; they are independent when $P(A \cap B) = P(A)P(B)$. If both have positive probability, mutual exclusivity makes them dependent.

Use the multiplication principle for successive choice counts. Use $P(n,r)$ for ordered selections and $C(n,r)$ for unordered selections, with $0 \leq r \leq n$. Counting favorable divided by total outcomes is valid only when elementary outcomes are equally likely.

Checked illustration. A code uses one of 4 letters followed by two distinct digits. There are $4 \times 10 \times 9 = 360$ codes. Choosing 2 digits as an unordered pair would instead use $C(10,2)=45$, which answers a different question.

Session 3: conditional, joint, marginal, and total probability

Conditioning changes the denominator to the event after the bar. A joint probability describes co-occurrence; a marginal totals across other categories.

Session 3: condition, multiply, then add paths

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad P(B) > 0$$

$$P(A \cap B \cap C) = P(A)P(B | A)P(C | A \cap B)$$

$$\text{Partition } B_i: P(A) = \sum P(B_i)P(A | B_i)$$

The denominator requirement $P(B) > 0$ matters. For a tree, multiply along a branch and add mutually exclusive branches. Total probability requires a partition: nonoverlapping branches that cover S .

Derivation. If partition events are $B_1, \dots, B_k$, then $A = (A \cap B_1) \cup \dots \cup (A \cap B_k)$. The pieces are disjoint, so $P(A) = \sum P(A \cap B_i) = \sum P(B_i)P(A | B_i)$.

Checked illustration. A delivery comes by air with probability 0.3 and road with 0.7. Late probabilities are 0.1 and 0.2. Thus $P(\text{late}) = 0.3(0.1) + 0.7(0.2) = 0.17$, while $P(\text{air} | \text{late}) = 0.03 / 0.17 \approx 0.1765$.

Session 4: Bayes theorem and naive Bayes

Bayes reverses a condition. The numerator is one prior-times-likelihood path; the denominator sums all paths to the evidence.

$$P(B_j | A) = \frac{P(B_j)P(A | B_j)}{\sum_{i=1}^k P(B_i)P(A | B_i)}$$

$P(C_j)$ is a prior, $P(x | C_j)$ a likelihood, $P(x)$ the evidence, and $P(C_j | x)$ the posterior. Naive Bayes assumes features are independent conditional on the class:

$$P(x_1, \dots, x_m | C_j) = \prod_i P(x_i | C_j)$$

For classification, compare $P(C_j) \prod P(x_i | C_j)$ because the evidence is shared. Normalize by the sum of class scores to report posteriors. With category count n_i , total count N , V possible categories, and smoothing strength $\alpha > 0$, additive smoothing is $(n_i + \alpha) / (N + \alpha V)$; adding α to every numerator adds αV to the denominator. Log scores turn products into sums and reduce underflow. Conditional independence can be unrealistic, and naive Bayes probabilities may be poorly calibrated even when classification is useful.

Checked illustration. Priors are 0.7, 0.3 and an observed feature has likelihoods 0.1, 0.6. Scores are 0.07, 0.18; evidence is 0.25; posteriors are 0.28, 0.72.

Session 5: random variables and joint distributions

A random variable maps each outcome to a number. A discrete PMF assigns point masses; a continuous PDF assigns density, with probabilities obtained from areas. The CDF $F(x)=P(X\leq x)$ works for both and never decreases.

Session 5: mass, density, moments, and joints

PMF: $p(x) \geq 0$, $\sum p(x)=1$ PDF: $f(x) \geq 0$, $\int f(x)dx=1$

$E[X] = \sum xp(x)$ or $\int xf(x)dx$ $\text{Var}(X)=E[X^2]-E[X]^2$

$p_x(x) = \sum_y p_{x,y}(x,y)$ $p(x|y) = \frac{p(x,y)}{p_y(y)}$

Independence requires joint factorization; zero covariance alone does not.

Expectation and variance require the relevant sums or integrals to exist; finite variance also requires finite mean. A joint distribution describes (X,Y) . Sum or integrate over y to obtain the X marginal. Divide a joint by a positive marginal to obtain a conditional distribution. Independence requires factorization for all supported pairs. It implies zero covariance when moments exist, but the converse fails: for symmetric X , X and X^2 can have zero covariance while being dependent.

In the compact notation $\int [a,d]$, the bracketed values are the lower and upper limits of integration. For the simple density $f(x)=cx$ on $0\leq x\leq b$, use $\int x dx=x^2/2$ and $\int x^2 dx=x^3/3$. Normalization gives $1=\int [0,b] cx dx=cb^2/2$, so $c=2/b^2$. Interval probability is area: $P(a\leq X\leq d)=\int [a,d] cx dx=c(d^2-a^2)/2$. Expectation adds one factor of x : $E[X]=\int [0,b] x(cx)dx=c\int [0,b] x^2dx=cb^3/3$.

Checked illustration. If $P(X=0,1,2)=(0.25,0.50,0.25)$, then $E[X]=1$, $E[X^2]=1.5$, and $\text{Var}(X)=1.5-1^2=0.5$.

Session 6: named distributions

Session 6: match the data-generating story

Bernoulli: $x \in \{0,1\}$, $E[X]=p$, $\text{Var}(X)=p(1-p)$

Binomial: $P(X=k)=C(n,k)p^k(1-p)^{n-k}$, $k=0,\dots,n$

Poisson: $P(X=k)=e^{-\lambda} \frac{\lambda^k}{k!}$ $k=0,1,\dots$; $E[X]=\text{Var}(X)=\lambda$

Normal: $Z = \frac{X-\mu}{\sigma}$ support all real, $\sigma > 0$

t: all real; chi-square and F: nonnegative.

Bernoulli models one binary trial. Binomial counts successes in fixed n independent trials with constant success probability p . Poisson counts events in a fixed exposure under a constant-rate, independent-increments model; rescale λ with exposure. Normal is continuous and symmetric, with every real value possible.

Write a binomial count as $X=I_1+\dots+I_n$, where each Bernoulli indicator has $E[I_i]=p$ and $\text{Var}(I_i)=p(1-p)$. Linearity gives $E[X]=np$; independence makes variances add, so $\text{Var}(X)=np(1-p)$.

Student's t has all-real support and heavier tails than normal; its degrees of freedom matter. A chi-square variable is a sum of squares of independent standard normals and has nonnegative support. An F variable is a ratio of independent chi-square variables after scaling by their degrees of freedom and is nonnegative. These are introductions only; hypothesis testing is outside Midsem 1.

Checked illustration. For $X\sim\text{Poisson}(1.5)$, $P(X=0)=e^{-1.5}\approx 0.2231$, and both mean and variance are 1.5.

Session 7: sampling, the CLT, and confidence intervals

Simple random sampling assigns equal selection probability to each size- n subset. Stratified sampling partitions the population and samples within every stratum; correct weights matter if sampling fractions differ. A sampling distribution is the repeated-sample distribution of a statistic.

Session 7: repeated-sample behavior

$$E[X]=\mu \quad \text{Var}(X)=\frac{\sigma^2}{n} \quad SE(X)=\frac{\sigma}{\sqrt{n}}$$

$$\text{CLT: } \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \text{ approaches } N(0,1)$$

$$\text{Known } \sigma: \bar{X} \pm z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \text{Unknown } \sigma: \bar{X} \pm t_{1-\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

Coverage belongs to the method; a realized frequentist interval assigns no probability to fixed μ .

For independent identically distributed observations with finite variance, $E[X]=\mu$, $\text{Var}(X)=\sigma^2/n$, and $SE(X)=\sigma/\sqrt{n}$. The CLT says the standardized mean approaches a standard normal distribution as n grows. It does not make the original observations normal and supplies no universal $n=30$ guarantee. A literal SRS without replacement from a finite population is dependent: for sampling fraction n/N , multiply the SE by the finite-population correction $\sqrt{[(N-n)/(N-1)]}$. The uncorrected formulas instead use an iid/superpopulation interpretation or a negligible sampling fraction.

For known σ , a mean interval is $\bar{x} \pm z^* \sigma / \sqrt{n}$. For unknown σ , use $\bar{x} \pm t^* s / \sqrt{n}$ with $n-1$ degrees of freedom. With independent observations from a normal population, this t interval is exact even for small n ; for a nonnormal population it is a large-sample approximation justified by suitable CLT conditions, not an exact small-sample result. In repeated sampling, about the stated percentage of intervals from the procedure cover fixed μ ; the realized frequentist interval does not assign probability to μ .

Checked illustration. With $n=64$, $\bar{x}=52$, known $\sigma=8$, and $z^*=1.96$, $SE=1$, margin 1.96, and the 95% interval is (50.04, 53.96).

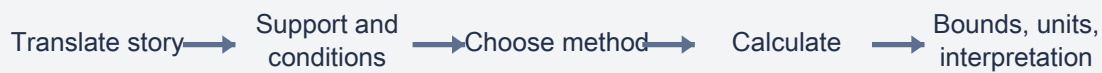
Extra worked examples

Each example teaches a different model or interpretation check.

- Binomial calculation: if 10 independently inspected items have constant defect probability 0.1, then $P(X=2)=C(10,2)(0.1)^2(0.9)^8 \approx 0.1937$; $E[X]=1$, which is a long-run average and need not be an observed integer.
- Poisson calculation and exposure: at $\lambda=3$ requests per minute, $P(X=0)=e^{-3} \approx 0.0498$. Two minutes uses $\lambda=6$, not 3, when the constant-rate process assumptions hold.
- Normal standardization: for a normal completion time with $\mu=50$ seconds and $\sigma=5$ seconds, 60 seconds has $z=(60-50)/5=2$.
- Unknown-sigma interval: with $n=25$, $\bar{x}=40$, $s=5$, and supplied $t^*=2.064$ for 24 degrees of freedom, $SE=1$, the margin is 2.064, and the interval is (37.936, 42.064). Its 95% claim concerns long-run procedural coverage.

Session 8: integrated review

Session 8 adds no theory. Use this decision chain:



- 1 Translate the story into a population, sample, events, or random variable.
- 2 State support, units, denominator, sampling design, and assumptions.
- 3 Choose the smallest valid rule.
- 4 Show formula, substitution, and non-obvious steps.
- 5 Check probability bounds, normalization, nonnegative variance, support, and unit scale.
- 6 Interpret in the original context without reversing a condition or overclaiming a confidence interval.

Checked illustration. If an event has daily Poisson rate 3 and the stated process model is valid, a two-day count uses $\lambda=6$, not 3; its support remains the nonnegative integers and its expected count is 6.

Compact mistake checklist

- Did I state the quartile convention and distinguish fences from whiskers?
- Did I use $\emptyset$, and did I distinguish exclusive from independent?
- Is the event after | my denominator?
- Did I normalize Bayes scores if probabilities were requested?
- Did I distinguish mass from density and check moment existence?
- Does the chosen distribution match its story and support?
- Did I distinguish individual spread from estimator standard error?
- Did I state units, bounds, conditions, and interpretation?

Conceptual Drills - Answer Free

Answer each in complete sentences. State any condition that makes the claim valid.

Session 1

C01 - Population language

A factory studies the lifetimes of every lamp it will produce next month by testing 80 lamps. Identify the population, sample, parameter of interest, and corresponding statistic.

C02 - Robust summaries

Compare mean with median and standard deviation with IQR when a dataset contains one extreme high value.

C03 - Bounds and shapes

Contrast the empirical rule with Chebyshev's theorem, including shape and variance conditions and the type of conclusion each supplies.

Session 2

C04 - Event structure

For one fair die roll, distinguish an outcome, an elementary event, a composite event, the sample space, and the empty event.

C05 - Exclusive or independent

Explain why two events with positive probability cannot be both mutually exclusive and independent.

C06 - Ordered selection

Compare permutations and combinations by describing one selection where changing order creates a new result and one where it does not.

Session 3

C07 - Conditional denominator

Explain why $P(A | B)$ uses $P(B)$ in its denominator and state when this expression is defined.

C08 - Table language

Distinguish joint, marginal, and conditional probability for two categorical variables in a two-way table.

C09 - Total probability

State the two conditions that a collection of branches must satisfy before the law of total probability may add their paths.

Session 4

C10 - Bayes vocabulary

Define prior, likelihood, evidence, and posterior for a class C and observed data x .

C11 - Naive assumption

State the exact naive Bayes independence assumption and explain how it differs from unconditional feature independence.

C12 - Scores and probabilities

Explain when unnormalized naive Bayes scores are sufficient, when normalization is required, and why log scores may be safer numerically.

Session 5

C13 - Mass and density

Compare a PMF, PDF, and CDF, including their normalization and the meaning of $P(X=x)$ for a continuous variable.

C14 - Moments

Define expectation and variance, state the useful computational variance identity, and give the existence condition.

C15 - Dependence

Explain the exact relationship between independence and zero covariance, including the needed moment condition.

Session 6

C16 - Bernoulli and binomial

Contrast a Bernoulli random variable with a binomial count and list every binomial trial condition.

C17 - Binomial and Poisson

Compare supports, parameters, means, variances, and data-generating stories for binomial and Poisson variables.

C18 - Continuous sampling families

Compare the support and qualitative shape of normal, Student's t, chi-square, and F distributions, and state why degrees of freedom matter.

Session 7

C19 - Sampling designs

Compare simple random and stratified sampling, including one reason to stratify and one weighting caution.

C20 - Three distributions

Distinguish a population distribution, the observed distribution in one sample, and the sampling distribution of the sample mean.

C21 - CLT scope

State what the central limit theorem approaches, its central assumptions, and why $n=30$ is not a universal rule.

Session 8

C22 - Confidence language

Give a correct frequentist interpretation of a 95% confidence procedure and reject the common probability-about-fixed-parameter interpretation.

C23 - Method-selection audit

List, in order, the checks you would perform before and after a probability or statistics calculation.

C24 - Unit and support checks

Explain the expected units of probability, mean, standard deviation, variance, and standard error, and why support checks catch modeling errors.

Additional Session 7 practice

C25 - Skewness and CLT convergence

Explain why a highly skewed or heavy-tailed population may require a larger sample before the sampling distribution of the mean is well approximated by a normal distribution.

C26 - Confidence procedure versus realized interval

Distinguish a 95% confidence procedure from the claim that a fixed parameter has 95% probability of lying in one realized frequentist interval.

Numerical Drills - Answer Free

For every item, show the governing formula, supplied-value substitution, non-obvious steps, units or domain and validity conditions, and interpretation.

Session 1

N01 - Center and spread

For ordered measurements 4, 6, 7, 9, 14, calculate the mean, median, population variance, and population standard deviation.

N02 - Weighted mean

A final score uses assignments 30%, project 25%, and exam 45%. Scores are 84, 76, and 91. Calculate the weighted mean and interpret its unit.

N03 - Quartiles and fences

For ordered data 2, 3, 5, 8, 9, 11, 20, use the median-of-halves convention that excludes the overall median. Find the five-number summary, IQR, both fences, and any flagged values.

Session 2

N04 - Inclusion-exclusion

In a group, $P(A)=0.55$, $P(B)=0.40$, and $P(A \cap B)=0.18$. Find $P(A \cup B)$ and the probability of neither event.

N05 - Product rule

A badge consists of one of 5 symbols, followed by one of 4 colors, followed by a three-digit string whose digits may repeat. Count the badges.

N06 - Ordered versus unordered

From 10 finalists, count (a) three-person committees and (b) gold-silver-bronze assignments. State why the counts differ.

Session 3

N07 - Conditional table

Among 150 orders, 90 are online and 24 of those are returned; 60 are in-store and 6 of those are returned. Find $P(\text{return})$, $P(\text{online}|\text{return})$, $P(\text{online}|\text{return})$, and $P(\text{return}|\text{online})$.

N08 - Sequential draws

A bag contains 5 blue and 3 gold tokens. Two are drawn without replacement. Find the probability of two gold tokens and the probability of at least one gold token.

N09 - Total probability

A message uses channel A with probability 0.4 and channel B with probability 0.6. Failure probabilities are 0.03 and 0.08. Find the overall failure probability and $P(B|\text{failure})$.

Session 4

N10 - Two-cause Bayes

A component comes from line L1 with probability 0.65 and L2 with 0.35. Defect probabilities are 0.02 and 0.06. Given a defect, find the posterior probability it came from L2.

N11 - Naive Bayes scores

Classes R and S have priors 0.55 and 0.45. Given R, binary features $x_1=1$ and $x_2=1$ have probabilities 0.7 and 0.4; given S, they have probabilities 0.3 and 0.8. Under the naive assumption, calculate both scores, normalized posteriors, and predicted class.

N12 - Add-one smoothing

In class C, 12 training documents contain 30 token positions over a vocabulary of 8 words. A word occurs zero times. Using add-one smoothing, calculate its conditional probability. Also calculate the estimate without smoothing and explain the product-score consequence.

Session 5

N13 - Normalize a PMF

For $x=0,1,2,3$, let $p(x)=k(x+1)$. Find k , the CDF value $F(1)$, $E[X]$, and $\text{Var}(X)$.

N14 - Continuous density

Let $f(x)=cx$ for $0 \leq x \leq 2$ and zero otherwise. Find c , $P(0.5 \leq X \leq 1.5)$, and $E[X]$.

N15 - Joint PMF

A joint PMF has rows $X=0,1$ and columns $Y=0,1$ with masses 0.10, 0.20 in the first row and 0.30, 0.40 in the second. Find both marginals, $P(X=1|Y=0)$, $E[X]$, $E[Y]$, and test independence.

Session 6

N16 - Binomial count

For $X \sim \text{Binomial}(7, 0.2)$, find $P(X=2)$, $P(X \geq 1)$, the mean, and variance.

N17 - Poisson exposure

Calls follow a Poisson-process model with mean 1.2 per 15 minutes. For 45 minutes, find the new λ , $P(X=0)$, $P(X \geq 1)$, mean, and standard deviation.

N18 - Normal standardization

A valid normal model has mean 100 mm and standard deviation 4 mm. Convert 94 and 106 mm to z-scores and, using $P(-1.5 \leq Z \leq 1.5) = 0.8664$, find $P(94 \leq X \leq 106)$.

Session 7

N19 - Sampling mean moments

Independent observations have population mean 30 g and standard deviation 12 g. For samples of size 36, find $E(\bar{X})$, $\text{Var}(\bar{X})$, and $\text{SE}(\bar{X})$ with units.

N20 - Known-sigma interval

An independent sample has $n=81$, $\bar{x}=44.0$ minutes, and known population standard deviation 9.0 minutes. Using $z^*=1.96$, construct and interpret a 95% confidence interval for the population mean.

N21 - Unknown-sigma interval

An independent normal-population sample has $n=16$, $\bar{x}=25.0$ mg, and $s=4.0$ mg. Using the supplied 95% critical value $t^*=2.131$ with 15 degrees of freedom, construct and interpret the interval.

Session 8

N22 - Select and calculate

Eight independent devices each fail during warranty with constant probability 0.1. Define the count variable, select a distribution, find the probability of no failures, and check support and conditions.

N23 - Bounds audit

A distribution has mean 50, standard deviation 6, and otherwise unknown shape. Use Chebyshev to give the guaranteed lower proportion between 38 and 62. State why the empirical rule is not justified by the given information.

N24 - Full workflow

A population has known standard deviation 15 units. An independent random sample of 100 has mean 62 units. Using $z^*=1.645$, construct a two-sided 90% confidence interval, check its width and units, and give the frequentist interpretation.

Additional Sessions 6 and 7 practice

N25 - Fair-die success count

Identify the distribution and its parameters for the number of sixes in 20 independent fair-die rolls. State the support and every model condition.

N26 - Poisson moments and zero count

A Poisson count has mean 4. Find its variance and $P(X=0)$, showing the formula and numerical value.

N27 - Standard error from population spread

Independent observations have population standard deviation 12. For samples of size 36, find the standard error of the sample mean and state its units relative to the observations.

Historical-Source-Aware BITS Practice - Answer Free

These are independently rewritten practice items. None is claimed to reproduce an authentic paper verbatim.

H01 - Union and conditioning

Classification: Authored BITS-style

Target course: AMLCS ZC418 Introduction to Statistical Methods

Institution: Not applicable - authored practice

Historical course code: Not applicable - authored practice

Historical term/year: Not applicable - authored practice

Source: Not applicable - authored practice

For events A and B , $P(A)=0.48$, $P(B)=0.37$, and $P(A \cup B)=0.68$. Calculate $P(A \cap B)$ and $P(A|B)$, stating the denominator condition.

H02 - Recover binomial parameters

Classification: Authored BITS-style

Target course: AMLCS ZC418 Introduction to Statistical Methods

Institution: Not applicable - authored practice

Historical course code: Not applicable - authored practice

Historical term/year: Not applicable - authored practice

Source: Not applicable - authored practice

A binomial variable has mean 4.8 and standard deviation $\sqrt{1.92}$. Infer n and p , then state the support and trial conditions.

H03 - Normalize a PMF

Classification: Authored BITS-style

Target course: AMLCS ZC418 Introduction to Statistical Methods

Institution: Not applicable - authored practice

Historical course code: Not applicable - authored practice

Historical term/year: Not applicable - authored practice

Source: Not applicable - authored practice

For $x=1,2,3,4$, a proposed PMF is $p(x)=c(5-x)$. Find c , the CDF value $F(2)$, and $E[X]$, and check validity.